BAPA-Net: Cross-Domain Few-Shot Object Detection in Remote Sensing via Background-Aligned Prototype Aggregation

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***Abstract*—While large-scale supervised learning has catalyzed remarkable strides in remote sensing object detection (RSOD), its real-world efficacy is frequently stifled by the formidable costs of expert annotation and the inherent scarcity of high-quality samples. Cross-domain few-shot object detection (CD-FSOD) has emerged as a strategic paradigm to alleviate this dependency by facilitating knowledge transfer from label-rich natural domains to data-starved remote sensing environments. However, the pronounced geospatial domain shifts and intricate background interference often lead to suboptimal generalization in existing frameworks. To address these impediments, we introduce the Background-Aligned Prototype Aggregation Network (BAPA-Net), a consolidated framework meticulously crafted to enhance the invariance of cross-domain representations. Specifically, BAPA-Net integrates three pivotal mechanisms: 1) Cross-Domain Background Fusion Alignment (CBFA), which decouples and synchronizes source–target background distributions through an innovative sampling–clustering–aggregation pipeline, effectively mitigating low-level feature interference; 2) Soft-Mask-Guided Prototype Augmentation (SMPA), which employs dynamic probabilistic Gaussian masks to construct a continuous spatial attention mechanism, thereby filtering peripheral noise and refining prototype representativeness; and 3) Prototype Retrieval and Knowledge Aggregation (PRKA), which leverages a cross-domain Cache bank and attention-weighted retrieval to aggregate multisource information adaptively, ensuring persistent exploitation of limited labeled samples. Extensive evaluations across diverse benchmarks, including high-altitude, underwater, and deep-space scenarios, demonstrate that BAPA-Net consistently outperforms state-of-the-art (SOTA) methodologies. Specifically, our framework achieves substantial performance gains, improving the mAP by 5.0% to 10.8% relative to the baseline across 1, 5, and 10-shot protocols. Our results validate the framework's robustness in bridging the semantic gap between natural and remote sensing domains. The code is open source: https://github.com/JackLeeAI/BAPA-Net**

***Index Terms*—Background alignment, cross-domain few-shot learning (FSL), domain adaptation, remote sensing object detection (RSOD).**

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I. Introduction

REMOTE sensing object detection (RSOD) is crucial for global geospatial intelligence. By extracting precise target information from diverse platforms—including satellites, aircraft, and UAVs—it facilitates critical applications in national security and resource management. However, the practical deployment of RSOD is inherently constrained by the scarcity of high-quality, expert-annotated data and the prohibitive costs associated with large-scale labeling. Although traditional deep learning paradigms perform well under data-rich conditions, their performance often sharply deteriorates when confronted with typical complex, multiscale, and highly dynamic scene features in real-world remote sensing environments. To mitigate such reliance, few-shot learning (FSL) [1], [2] has surfaced as a compelling paradigm, exploiting prior knowledge to generalize to novel categories with minimal supervision.

The landscape of remote sensing FSL is predominantly bifurcated into metalearning paradigms [3], [4] and transfer-learning-based fine-tuning strategies [5], [6], [7]. These methodologies typically operate under the idealized premise that source and target domains are independent and identically distributed (i.i.d.). However, in the cross-domain few-shot object detection (CD-FSOD) context, these assumptions fail because the inherent domain differences in it stem from different sensor modalities and atmospheric conditions, which seriously weaken the discriminative ability of the pretrained representations. CD-FSOD navigates this challenge by

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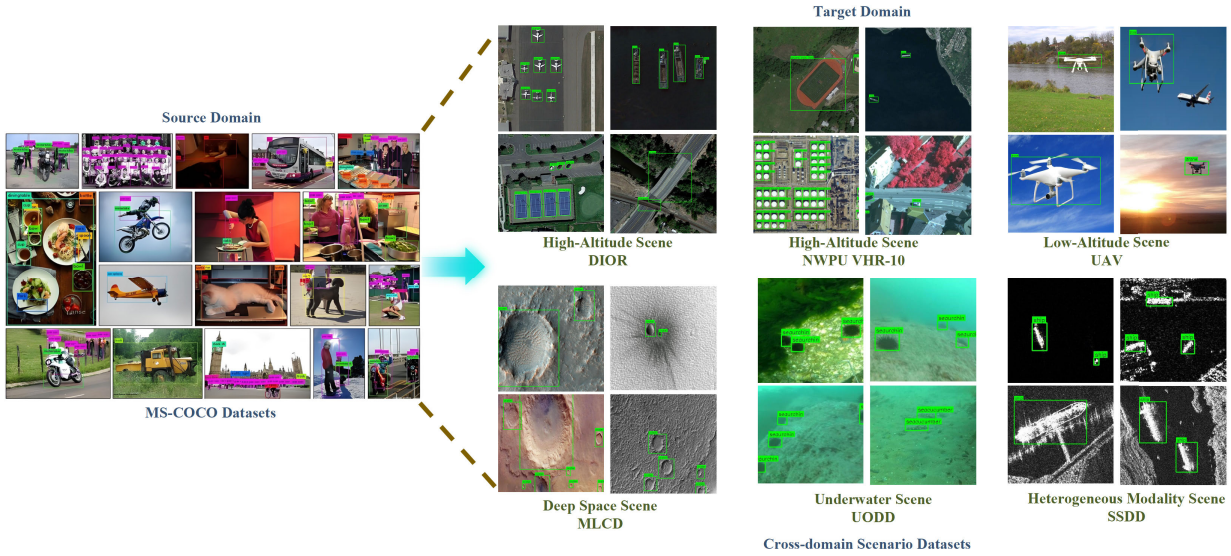


Fig. 1. CD-FSOD framework is optimized on the source corpus (MS-COCO) and subsequently validated across heterogeneous remote sensing target domains.

transferring latent knowledge from a label-rich source domain (such as natural images, for example) to a specific remote sensing target domain (as shown in Fig. 1). However, this transition is inherently hindered by substantial data heterogeneity; natural and remote sensing domains differ fundamentally in imaging perspective (nadir versus horizontal), object scales (multiscale versus uniform), and background complexity (cluttered geospatial versus structured natural scenes). To ensure effective transfer learning, it is imperative to decouple domain-specific noise from domain-agnostic semantics, allowing the model to "bridge" the gap by aligning background statistics and purifying foreground representations. Decoupling enables the model to focus on invariant structural features (e.g., aircraft geometry and ship contours) consistent across diverse remote sensing domains, while filtering domain-specific clutter (nadir views, speckle noise, artifacts, and complex backgrounds). This prevents erroneous associations between object semantics and irrelevant source-domain attributes (e.g., airplanes with blue skies in MS COCO), avoiding negative transfer and promoting robust cross-domain generalization. Pretrained weights from natural datasets provide a strong foundation due to shared low-level primitives (edges, textures, and shapes). Our decoupling mechanism ensures only these transferable structural insights are inherited, suppressing domain biases for stable transfer learning even in extreme cross-domain scenarios.

The current CD-FSOD research mainly explores feature alignment and prototype optimization. However, these methodologies face fundamental limitations. Traditional feature alignment usually tries to match the whole data distribution. Because of this, it often misses the small, specific details needed to find crowded targets in remote sensing images. Most prototype-based frameworks only focus on foreground enhancement, while ignoring the background interference features that are ubiquitous in complex geographical spatial environments. Existing methods rarely utilize the background statistical information of the target domain to actively counteract the domain shift at the background level. Therefore, the model is still very susceptible to feature interference, where irrelevant environmental noise is mistakenly regarded as semantic signals, resulting in significant localization deviations.

The recent proliferation of large-scale vision–language models, most notably contrastive language–image pretraining (CLIP), has brought about new dimensions of complexity. CLIP is a foundational vision–language model that learns transferable visual representations from natural language supervision by aligning images and their corresponding textual descriptions in a shared embedding space. There is growing evidence suggesting that the observed improvement in generalization effects might be an illusory phenomenon caused by data contamination during the large-scale pretraining process [8], [9]. To ensure a strict and unbiased assessment of true cross-domain adaptability, we strictly follow the closed-world evaluation protocol established by CD-ViTO [10]. By skipping the basic class pretraining for the target domain and relying solely on a handful of novel-class exemplars for adaptation, this protocol effectively isolates algorithm generalization and reflects the true scarcity within practical deployment scenarios.

To overcome these fundamental bottlenecks, we propose Background-Aligned Prototype Aggregation Network (BAPA-Net), a synergistic framework designed to achieve robust cross-domain few-shot detection through collaborative background alignment, instance purification, and knowledge inheritance. Unlike existing methods that treat domain adaptation and FSL as isolated processes, BAPA-Net integrates distribution-level background fusion with patch-level historical knowledge retrieval. This holistic approach ensures that the model can extract discriminative semantic signals while filtering out pervasive environmental noise, thereby establishing a stable distributional anchor for object detection across diverse and extreme remote sensing scenarios.

Comprehensive empirical evaluations performed across six distinct categories of remote sensing datasets substantiated

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the architectural advantage of BAPA-Net. Both qualitative and quantitative results demonstrated that our framework consistently outperformed the SOTA baseline models, with an average precision (mAP) that was significantly improved by 5.0% to 10.8% under challenging test protocols involving 1, 5, and 10 shots.

The principal contributions of this work are summarized as follows.

1) We propose a novel Cross-Domain Background Fusion Alignment (CBFA) module that employs an optimal transport mechanism to align cross-domain background statistics, providing a stable distributional anchor for FSL in complex environments.
2) We develop a Soft-Mask-Guided Prototype Augmentation (SMPA) module that introduces dynamic soft Gaussian masks for prototype purification, effectively filtering out nondiscriminative background noise from sparse support instances.
3) We design a Prototype Retrieval and Knowledge Aggregation (PRKA) module based on a patch-level cross-attention retrieval strategy, which aggregates multisource knowledge from a cross-domain cache bank to enhance the robustness of query detection.
4) Extensive experiments on multiple cross-domain remote sensing benchmarks demonstrate that BAPA-Net achieves state-of-the-art (SOTA) performance, particularly in 1-shot and 5-shot scenarios.

II. Related Work

A. Remote Sensing Object Detection

Contemporary remote sensing detection methodologies are predominantly stratified into convolutional neural networks (CNNs) and Transformer-based architectures. CNN-based detectors use convolution to get image features. This represented the dominant methodology in prior literature. Conventionally, these architectures are bifurcated into two-stage and single-stage paradigms. Two-stage methodologies, exemplified by Faster R-CNN [11] and Mask R-CNN [12], prioritize precision by initially deploying a Region Proposal Network (RPN) to identify prospective regions of interest, followed by a refinement phase for bounding box regression and classification. Subsequently, they refine the bounding boxes and classify the object proposals. Although this two-stage paradigm typically yields superior localization accuracy, it incurs high computational costs and significant latency, limiting its applicability in real-time scenarios. Conversely, single-stage architectures, notably the YOLO series [13], bypass the intermediate proposal generation phase to concurrently regress bounding coordinates and predict class probabilities directly from feature representations. This streamlined mechanism inherently fosters superior computational efficiency and accelerated inference velocities, rendering them particularly advantageous for latency-sensitive deployment scenarios such as UAV-based surveillance. Recently, detectors based on Transformer have attracted significant attention in the remote sensing field. Compared with convolutional designs, the Transformer-based designs utilize long-range dependencies and dynamic weight adjustments to promote more excellent global relationship modeling. This flexible structure helps solve common problems in remote sensing. It works well when objects are spread far apart or when their sizes change a lot. Methods like DETR [14], Deformable DETR [15], and DINO use global information to deal with messy object layouts. Recently, DINOv3 [16] further advanced this paradigm by introducing a simplified self-supervised formulation that scales effectively across diverse visual domains. They demonstrate robust performance in complex scenarios, such as large-scale aerial imagery or planetary surfaces with homogeneous backgrounds. Furthermore, these models employ an end-to-end design, eliminating the need for handcrafted components like nonmaximum suppression (NMS), thereby streamlining the detection pipeline. Although the CNN architecture is still widely favored due to its operational maturity and computational efficiency in time-critical remote sensing tasks, the Transformer framework offers unprecedented global representation depth. Consequently, this type of model provides a robust baseline for challenging scenarios, particularly in few-shot settings or cross-domain environments.

B. Few-Shot Object Detection

Methodologically, advancements within the FSOD landscape have primarily coalesced around four distinct research vectors: synthetic data augmentation, metalearning paradigms, fine-tuning-based transfer learning, and metric-centric relational modeling. The first strategy involves data augmentation. These methods employ generation and transformation techniques to expand the diversity of limited sample sets. Conventional geometric transformations, such as rotation and translation, are widely used to augment the data [17]. At the same time, models like GANs and diffusion models [18] can make new, realistic examples that look like the original ones. Within the specialized domain of remote sensing imagery analysis, few-shot feature augmentation methodologies [19] address the challenge of drastic scale variations. These techniques typically synthesize multiscale virtual instances to refine and optimize the underlying prototype distribution. The metalearning paradigm cultivates transferable optimization strategies via episodic task formulation, effectively internalizing the "learning-to-learn" heuristic within iterative metatrain/metatest loops [20]. A quintessential embodiment, model-agnostic metalearning (MAML), facilitates swift parameter recalibration, thereby empowering remote sensing architectures to achieve rapid convergence even when constrained by sparse aerial or satellite imagery [21]. In the transfer learning paradigm, "prior knowledge" is inherited by improving the base encoder (usually a feature extractor initialized from a wide source-domain corpus), matching its parameter space with the unique features of sparse new class instances, thereby promoting the robust transition of discriminative ability [5], [6], [7]. Metric learning works by creating a feature space where distances between points can be measured and improved [22], [23], [24], [25], [26], [27], [28]. The primary objective is to enhance intraclass compactness by minimizing the feature-space distance among samples that belong to the same category. At the same time, it tries to push

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different classes as far apart as possible to make the boundaries clear. As a common target detection and segmentation method, it prioritizes obtaining a transferable similarity metric, which can distinguish new classes based on relative feature proximity. Representative results include the following: Karlinsky et al. [22] use class representatives to construct a class distribution model; Leng et al. [23] enhance robustness through sampling invariance constraints; and Jiang et al. [24] use multiscale alignment to improve discrimination ability. At the same time, Vinyals et al. [29] and Snell et al. [30] establish the foundation of prototype classification. For remote sensing scenarios, Li et al. [31] improve the performance of new class detection through metric learning and knowledge inheritance, while Jiang et al. [32] alleviate overfitting in few-sample segmentation through metric alignment. Beyond advances in model design, real-world multiobject instance segmentation also depends critically on the reliability of multimodal inputs and the accuracy of their cross-modal alignment. For example, DADIGAN [33] and CIRSM-Net [34] mix SAR and optical data together. They use iterative registration to align the images, which creates a solid starting point for transferring semantic information.

C. Cross-Domain Object Detection

Cross-domain object detection primarily aims to mitigate domain shift by reducing the distribution discrepancy between the source and target domains [35]. To this end, current investigations have bifurcated into two distinctive methodological trajectories: domain adaptation and domain generalization. Within DA, adversarial learning [36] and statistical distance minimization [37] represent the two mainstream strategies for alleviating domain shift. Within the more formidable landscape of cross-domain few-shot detection, several advanced approaches have been proposed. For instance, DE-ViT [38] leverages large-scale source-domain pretraining to obtain global attention representations and fine-tunes the full Vision Transformer on the target domain, indirectly mitigating domain shift through long-range dependency modeling. CD-ViTO [10] learns global class prototypes during source pretraining and updates them via exponential moving average during few-shot fine-tuning to achieve class-level cross-domain prototype alignment; CDFormer [39] incorporates cross-domain attention and hierarchical feature fusion to capture semantic correlations across domains, achieving strong performance in cross-modal SAR-optical satellite imagery; The Instance Cache Fusion (IFC) framework [40] constructs a target-domain "instance cache" and dynamically injects semantic features from scarce labeled samples via a retrieval-and-update mechanism; Domain-RAG [41] generates domain-aligned background samples through retrieval-guided augmentation to enrich training data for CD-FSOD. Meanwhile, TSCCD [42] explores temporal self-construction for unsupervised cross-domain learning, while DHSNet [43] employs a dual-head self-training network to enhance cross-scene classification performance, validating the efficacy of self-supervised adaptation in hyperspectral domains. ETS [44] further introduces an augmentation-search strategy with foundation models to optimize cross-domain few-shot detection.

Despite these advances, current methods suffer from three key limitations: first, they largely neglect the background information embedded in the limited labeled samples of downstream few-shot tasks and lack dedicated modules to address the complex background interference inherent in remote sensing, making background-induced domain shift a persistent bottleneck; second, although some methods update prototypes online, they rely solely on the extremely scarce samples of the current task and lack systematic mechanisms to reuse historical cross-domain knowledge, leading to prototype noise accumulation under severe data scarcity; third, most studies focus on a single remote sensing scenario and fail to account for the diversity across aerial, satellite, and planetary imaging domains, resulting in a notable absence of truly universal cross-domain few-shot detection frameworks for the full spectrum of remote sensing data.

III. Methods

A. Problem Formulation

We leverage CD-FSOD to address a prevalent challenge in remote sensing: we often lack labels and face huge changes between different data environments. In this work, we want the model to take features it learned from a labeled source dataset ($\mathcal{D}_s = \{X_s, Y_s\}$) and use them in a target dataset ($\mathcal{D}_t = \{X_t, Y_t\}$) that has very few labels. This target data X_t comes from many places, such as deep space, high-altitude planes, drones, underwater areas, and SAR sensors. Our labels $Y_s = \left\{(b_{s,i}, c_{s,i})\right\}_{i=1}^{N_s}$ and $Y_t = \left\{(b_{t,j}, c_{t,j})\right\}_{j=1}^{N_t}$ use b for boxes and c for classes. Since the source classes $\mathcal{C}_s$ and target classes $\mathcal{C}_t$ do not overlap, the model must find objects it has never seen before. To see how well the model moves knowledge to the new domain, we follow the standard N-way K-shot protocol. In this setup, the target domain $\mathcal{D}_t$ has N new categories. Each category has only K labeled images, which we call the support set $S^{N\times K} \subset \mathcal{D}_t$. The model then looks at a query set $Q = X_{t,\text{query}} \subset \mathcal{D}_t$ and tries to predict the labels and bounding boxes $\hat{Y}_t = \{(\hat{b}_{t,j}, \hat{c}_{t,j})\}$. We usually test K at 1, 5, and 10. These different values of K help us understand how the model works when data is extremely scarce or just limited.

To ensure a rigorous and fair evaluation, we follow the closed-world protocol of CD-ViTO [10]. In particular, we avoid using large-scale general-purpose pretrained detectors (e.g., Grounding DINO [45]) as the infrastructure, because their pretraining corpora may implicitly contain remote sensing imagery, which can lead to inflated performance that does not reflect genuine cross-domain generalization. Under this constraint, the evaluation is less affected by potential external-data leakage and better isolates the intrinsic cross-domain adaptability of the proposed method. Different from the conventional pipeline that performs additional target-domain base-class pretraining before few-shot transfer, our setting does not rely on any target-domain base-class annotations. Instead, the detector is initialized from standard public pretraining (e.g., ImageNet/MS-COCO, consistent with DE-ViT/CD-ViTO) and is adapted in a task-specific manner using only the support instances in $S^{N\times K}$, with performance reported on the corresponding query set Q. This protocol better matches

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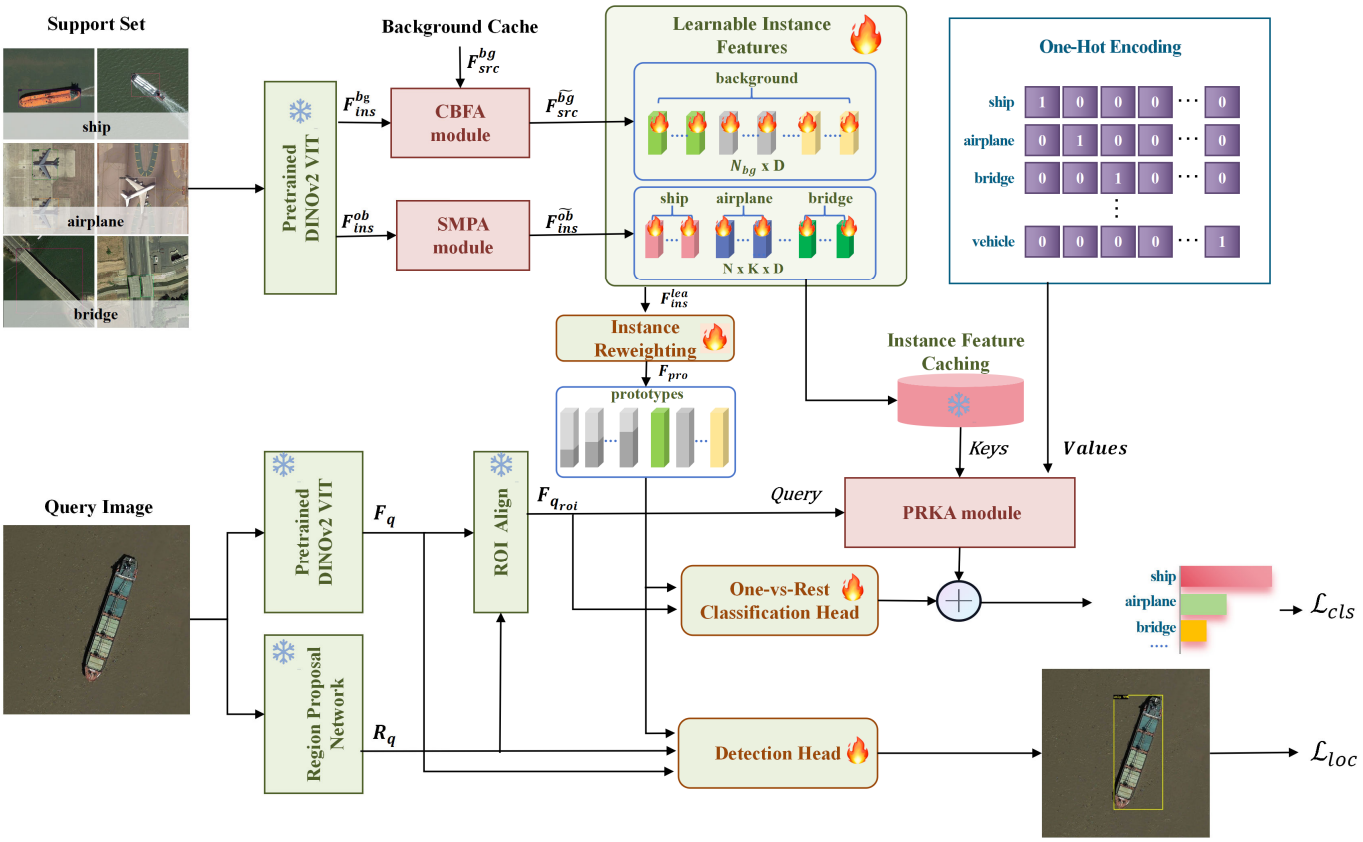


Fig. 2. Overall architecture of BAPA-Net. (Top) Prototype construction: support images are processed by DINOv2 ViT. CBFA aligns cross-domain backgrounds ($F^{\text{bg}}_{\text{src}}$), and SMPA refines instance features (F^{ob}_{ins}). These purified features are stored in the cache bank as keys, which are then integrated into learnable instance features and optimized via instance reweighting to form final prototypes (F_{pro}). (Bottom) Query inference: ROI features ($F_{q_{\text{roi}}}$) are enhanced by the PRKA module through patch-level cross-attention with the cache bank. The retrieved knowledge is fused with detection heads for classification ($\mathcal{L}_{\text{cls}}$) and localization ($\mathcal{L}_{\text{loc}}$). Snowflake and fire icons denote frozen and learnable parameters, respectively.

practical deployment, where adaptation must be carried out with only a handful of labeled samples in the target domain.

B. Overall Network

Fig. 2 presents an overview of the BAPA-Net architecture. It integrates the hybrid detection paradigm of Mask R-CNN [12] and Deformable DETR [15]. To overcome the heterogeneity between the natural source domain and diverse remote sensing domains, BAPA-Net implements a multilevel alignment strategy through three core components integrated into this hybrid architecture: the CBFA module, the SMPA module, and the PRKA module.

This design aims to combine the local inductive bias of CNNs with the inherent expansive global relationship modeling in Transformers. This architecture is specifically targeted at the three key challenges of CD-FSOD in the remote sensing field: the background–statistics mismatch between the source and target domains, prototype uncertainty induced by scarce samples, and fragmented knowledge arising from limited supervision. BAPA-Net is developed based on the open-source detector DE-ViT [38] and CD-ViTO [10]. The architecture of DE-ViT is supported by a pretrained DINOv2 ViT backbone network serving as a domain-agnostic feature extractor, which is further integrated with the RPN network. It is also enhanced through the RoI alignment mechanism and a dual-task branch consisting of specialized detection and classification heads. CD-ViTO optimizes DE-ViT by introducing learnable instance feature modules, instance weighting modules, and domain prompter modules. On this basis, BAPA-Net introduces three new core components: the CBFA module for explicit background alignment, the SMPA module for foreground purification, and the PRKA module for cross-domain knowledge retrieval, thereby constructing a complete and unified detection framework.

The operational procedure of BAPA-Net is bifurcated into two sequential phases: the initial distillation of support-set priors, followed by the execution of query-set object localized detection.

1) Support-Set Feature Processing: Taking a collection of support instances S as input, the DINOv2 backbone first obtains instance features $F_{\text{ins}} = \{F^{\text{ob}}_{\text{ins}}, F^{\text{bg}}_{\text{ins}}\}$. Specifically, $F^{\text{ob}}_{\text{ins}}$ captures the fine-grained foreground object representations containing target-specific semantics, while $F^{\text{bg}}_{\text{ins}}$ encodes the surrounding instance background features reflecting the environmental context. To address background domain shift, the CBFA module fuses and updates the source-domain background features $F^{\text{bg}}_{\text{src}}$ with the target-domain background features $F^{\text{bg}}_{\text{ins}}$ using optimal transport and an incremental update strategy. An adaptive weighted fusion unit then generates a

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unified background representation $\tilde{F}^{\text{bg}}_{\text{src}}$, achieving alignment of background distributions across domains. To mitigate prototype noise, the SMPA module refines the foreground features $F^{\text{ob}}_{\text{ins}}$: a localized 1×1 spatial mapping, modulated by a sigmoid nonlinearity, yields an adaptive soft-attention mask. This mask subsequently serves to recalibrate the support representations, facilitating the exclusionary filtering of fine-grained categorical details. Subsequently, a spatial average pooling layer aggregates these masked features into an enhanced initial object prototype $\tilde{F}^{\text{ob}}_{\text{ins}}$. This initial prototype splits into two branches: one is processed by CD-ViTO's learnable instance feature updater and instance reweighting module to yield the final feature prototype F_{pro}; the other is stored in a cross-domain prototype cache bank to enable knowledge reuse in subsequent tasks.

2) Query-Set Detection and Optimization: For the query image q, DINOv2 first generates the visual feature F_q. The RPN generates a set of candidate regions R_q, which are spatially normalized through the RoIAlign operation. This process converts them into fixed-sized RoI representations $F_{q_{\text{roi}}}$, facilitating subsequent downstream refinement processing. In the classification process, the representative feature prototype F_{pro} is integrated with the pooled $F_{q_{\text{roi}}}$ representation through the classification head. This interaction helps to derive the classification objective $\mathcal{L}_{\text{cls}}$, which aims to ensure consistency between the derived prediction distribution and the reference of the true label. In the localization task, the visual feature F_q, the region proposal R_q, and the feature prototype F_{pro} are jointly input into the detection head to generate the regression loss $\mathcal{L}_{\text{reg}}$. To guide the network to converge, the calculation method of the comprehensive training loss is as follows:

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{reg}} + \mathcal{L}_{\text{dp}} \tag{1}$$

where $\mathcal{L}_{\text{dp}}$ denotes the domain prompter loss adopted from the CD-ViTO baseline [10], whose effectiveness in facilitating domain-adaptive feature learning has been extensively validated. Specifically, $\mathcal{L}_{\text{dp}}$ is formulated as a composite objective: $\mathcal{L}_{\text{dp}} = \mathcal{L}_{\text{domain}} + \mathcal{L}_{\text{proto}} + \mathcal{L}_{\text{proto}_{cls}}$. Here, $\mathcal{L}_{\text{domain}}$ and $\mathcal{L}_{\text{proto}}$ are InfoNCE-based consistency losses that align features at the domain and prototype levels respectively, while $\mathcal{L}_{\text{proto}_{cls}}$ ensures categorical discriminability. By incorporating this established component, we maintain consistent optimization of the underlying backbone while focusing on our proposed modules.

To mitigate the accumulation of prototype noise in cases of extremely scarce samples, during the inference process, the ROI features $F_{q_{\text{roi}}}$ are passed to the PRKA module. In this module, they are retrieved through cosine similarity and matched with the prebuilt cross-domain prototype cache bank. The classification scores are calculated using one-hot encoding to assist the classification head in making category predictions, building an end-to-end pipeline that maps raw inputs directly to detection outputs. Through the design of the collaborative module, this pipeline effectively addresses the issues of background distribution bias and the limitation of a few samples, ensuring reliable detection performance in complex cross-domain remote sensing scenarios.

C. CBFA Module

To fully utilize the prior knowledge of background distribution in cross-domain remote sensing scenarios, the CBFA module was developed. This component utilizes entropy regularization for optimal transport plans to achieve feature alignment between different domain backgrounds. To robustly suppress noise during this process, we implement a multilevel strategy. First, we normalize features to eliminate scale discrepancies and incorporate domain shift penalties into the cost matrix to discourage matching between significantly disparate domains. Furthermore, we apply entropy regularization to smooth transport plans and prevent overfitting to local noise. The fusion process is further refined by employing conservative momentum fusion to preserve the stability of source priors. Leveraging these mechanisms, this module adaptively aligns and fuses target-domain background features with the source-domain background space, yielding a unified background representation that suppresses background-induced noise during object feature learning. Different from existing methods, which merely distinguish between foreground and background, CBFA introduces for the first time an adaptive fusion mechanism with distribution alignment, allowing background prototype weights to be updated online from just a few samples without retraining the full network.

Obtaining the background characteristics of the target domain $\mathbf{F}^{\text{bg}}_{\text{ins}}$ involves two key steps.

1) *Precise Background Region Separation:* The foreground and background are separated directly using bounding boxes with support for centralized annotation, without the need for an auxiliary segmentation model. Specifically, for each annotated support instance, pixels falling inside the bounding box coordinates are treated as foreground, whereas all pixels outside this box are assigned to the initial background distribution. Although there may be unmarked objects in the background in dense scenes, the global matching strategy of the optimal transport mechanism helps to mitigate the influence of these local outliers by focusing on the main background distribution. The backbone network of DINOv2 randomly extracts features from this nonforeground area as the original background features.
2) *Background Feature Clustering:* We apply K-means clustering to the raw background features extracted from the K-shot support set to obtain representative target-domain background prototypes. The number of clusters is selected by jointly considering the silhouette coefficient, the Davies–Bouldin index, and the Calinski–Harabasz score over a predefined search range, so as to balance cluster compactness and separation. This combined metric is evaluated within the predefined search space to ensure the stability and separability of the partitions. The resulting centroids initialize an online prototype learning framework, which optimizes the prototypes using momentum-based exponential moving average (momentum = 0.002) on L2 normalized features over 30 cycles. This process is executed once before each task for fine-tuning, and each support image

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takes approximately 2.8 ms. Let $\mathbf{F}_{\text{ins}}^{\text{bg}} = [\mathbf{f}_1^{\text{ins}}, \ldots, \mathbf{f}_{N_b}^{\text{ins}}]^\top \in \mathbb{R}^{N_b \times D}$ denote the background features obtained from K-shot instances in the target domain after clustering (N_b: number of cluster centroids; D: feature dimension), and let $\mathbf{F}_{\text{src}}^{\text{bg}} = [\mathbf{f}_1^{\text{src}}, \ldots, \mathbf{f}_{M_s}^{\text{src}}]^\top \in \mathbb{R}^{M_s \times D}$ represent the source-domain background prototypes precomputed on the source dataset (M_s: number of source prototypes). Both sets are first L2-normalized and mapped onto the unit hypersphere

$$\bar{\mathbf{F}}_{\text{src}}^{\text{bg}} = \text{Norm}\left(\mathbf{F}_{\text{src}}^{\text{bg}}\right), \quad \bar{\mathbf{F}}_{\text{ins}}^{\text{bg}} = \text{Norm}\left(\mathbf{F}_{\text{ins}}^{\text{bg}}\right). \tag{2}$$

To represent the background distributions of the dual domains, we compute the support-set centroids as follows:

$$\boldsymbol{\mu}_{\text{src}} = \frac{1}{M_s}\sum_{i=1}^{M_s} \bar{\mathbf{f}}_i^{\text{src}} \in \mathbb{R}^D, \quad \boldsymbol{\mu}_{\text{ins}} = \frac{1}{N_b}\sum_{j=1}^{N_b} \bar{\mathbf{f}}_j^{\text{ins}} \in \mathbb{R}^D. \tag{3}$$

Then, compute their Euclidean distance as a measure of domain shift

$$\delta_{\text{domain}} = \left\|\boldsymbol{\mu}_{\text{src}} - \boldsymbol{\mu}_{\text{ins}}\right\|_2. \tag{4}$$

Based on this, we construct a robust cross-domain cost matrix $\mathbf{C} \in \mathbb{R}^{M_s \times N_b}$ that integrates semantic similarity and global domain shift

$$\mathbf{C}_{ij} = \left(1 - \langle \bar{\mathbf{f}}_i^{\text{src}}, \bar{\mathbf{f}}_j^{\text{ins}} \rangle\right) \cdot (1 + \delta_{\text{domain}}) \tag{5}$$

where $\mathbf{C}_{ij}$ denotes the cost of transporting between the ith source background prototype and the jth target instance feature. This formulation ensures nonnegativity of the cost matrix, as both the cosine distance $(1 - \langle \cdot, \cdot \rangle)$ and the domain shift δ_{domain} are nonnegative. Specifically, the first term measures local semantic dissimilarity, while the second term, scaled by the global domain shift δ_{domain}, acts as an adaptive penalty: when the overall domain gap is large, this term increases the transport cost for dissimilar pairs, thereby discouraging risky matches and enforcing a more conservative alignment strategy. $\langle \cdot, \cdot \rangle$ denotes cosine similarity (i.e., inner product of L2-normalized vectors).

We adopt a uniform marginal distribution over background prototypes rather than weighting by cluster sizes to avoid bias toward frequently occurring background patterns in the support set. This ensures all learned generic background modes (e.g., sky, ground, and buildings) are treated equally, enhancing robustness to domain-specific imbalances and preventing overfitting to incidental statistics of the target domain: $\mathbf{a} = 1/M_s \mathbf{1}_{M_s}$, $\mathbf{b} = 1/N_b \mathbf{1}_{N_b}$. With these marginals, we address the entropy-regularized optimal transport problem

$$\min_{\boldsymbol{\Gamma} \in \Pi(\mathbf{a},\mathbf{b})} \langle \mathbf{C}, \boldsymbol{\Gamma} \rangle - \varepsilon H(\boldsymbol{\Gamma}) \tag{6}$$

where $\langle \mathbf{C}, \boldsymbol{\Gamma} \rangle = \sum_{i,j} \mathbf{C}_{ij} \boldsymbol{\Gamma}_{ij}$ denotes the Frobenius inner product representing the total transport cost, the coupling set is defined as $\Pi(\mathbf{a}, \mathbf{b}) = \{\boldsymbol{\Gamma} \in \mathbb{R}_+^{M_s \times N_b} \mid \boldsymbol{\Gamma} \mathbf{1}_{N_b} = \mathbf{a}, \boldsymbol{\Gamma}^\top \mathbf{1}_{M_s} = \mathbf{b}\}$, the entropy term is $H(\boldsymbol{\Gamma}) = -\sum_{i,j} \boldsymbol{\Gamma}_{ij} \log \boldsymbol{\Gamma}_{ij}$, and $\varepsilon > 0$ is the regularization coefficient. In order to establish a reliable correspondence between the source background prototype and the target background prototype, we employ the entropy-regularized optimal transport framework to calculate the matching matrix $\boldsymbol{\Gamma}^*$. Specifically, using the cost matrix $\mathbf{C}$ in (5) as the input, we efficiently obtain the transport plan $\boldsymbol{\Gamma}^*$ that satisfies the marginals $\mathbf{a}$ and $\mathbf{b}$ through the Sinkhorn algorithm. This step is not merely for minimizing the transport cost, but for obtaining matching weights with semantic awareness and domain robustness. Based on their matching confidence, each source background prototype will undergo adaptive updates

$$\tilde{\mathbf{F}}_{\text{ins}}^{\text{bg}} = (1 - \beta)\, \mathbf{F}_{\text{src}}^{\text{bg}} + \beta M_s \boldsymbol{\Gamma}^* \mathbf{F}_{\text{ins}}^{\text{bg}} \tag{7}$$

where $\boldsymbol{\Gamma}^*$ is the computed optimal transport plan, and β is a fusion weight used to control the contribution of the background knowledge of the target domain (fusion weight, $\beta = 0.4$, see Section IV-H). The updated background prototype set is represented as $\tilde{\mathbf{F}}_{\text{ins}}^{\text{bg}} = [\tilde{\mathbf{f}}_1^{\text{bg}}, \ldots, \tilde{\mathbf{f}}_{M_s}^{\text{bg}}]^\top \in \mathbb{R}^{N_{bg} \times D}$. This module realizes semantic perception, confidence guidance, and domain adaptation of background prototype fusion, effectively making the cross-domain background representation consistent while maintaining the integrity and generality of the source-domain background prior. The detailed pipeline and framework structure of the CBFA module are illustrated in Fig. 3.

D. SMPA Module

To enhance the recognition accuracy of instance prototypes when constructing from support samples in cross-domain few-shot scenarios, the SMPA module addresses the problem of prototype blurring caused by noise dilution. This module utilizes high-resolution foreground masks to generate spatially continuous soft attention weights, which serve to recalibrate the block-level features produced by the vision transformer. This effectively suppresses background noise while preserving the fine semantic details near the object boundaries.

In our setting, the target-domain supervision is provided in the form of bounding boxes. Therefore, we construct the binary foreground mask $\mathbf{M}$ by rasterizing each ground-truth box onto the $H \times W$ feature grid, setting pixels inside the box to 1 and those outside to 0. This design avoids introducing any external segmentation model and keeps SMPA fully lightweight, while still providing spatial cues that help suppress background contamination during prototype construction.

Crucially, this mechanism serves as an implicit foreground alignment strategy via feature purification. By filtering out domain-specific background noise within the object proposals, SMPA ensures that the extracted prototypes focus purely on the intrinsic semantic features of the objects (e.g., shape and structure), which are more invariant across domains than raw pixel features. Consequently, this enhances the transferability of object representations and reduces the intraclass variance caused by domain shifts.

Specifically, the representation of the foreground target takes the form of a feature tensor $\mathbf{F}_{\text{ins}}^{\text{ob}} \in \mathbb{R}^{L \times D}$, which is extracted from the support image. The spatial resolution is transformed into $L = H \times W$ elements (H, W are height and width, D is feature dimension). Additionally, we introduce a binary foreground mask $\mathbf{M} \in \{0, 1\}^{H \times W}$ to supply fine-grained, pixel-level spatial cues. To obtain spatially smooth attention weights, we convolve $\mathbf{M}$ with a Gaussian filter with standard deviation σ

$$\widetilde{\mathbf{M}} = \frac{G_\sigma(\mathbf{M})}{\max(G_\sigma(\mathbf{M}))} \tag{8}$$

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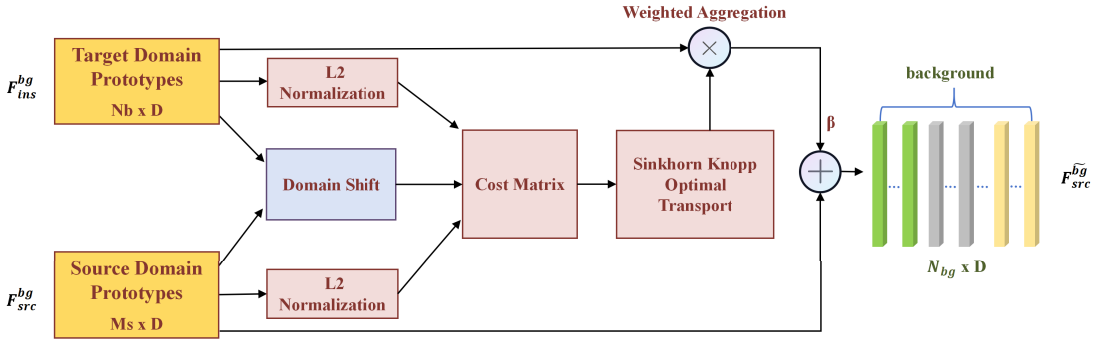


Fig. 3. Detailed framework structure of the CBFA module, illustrating the optimal transport-based background distribution alignment process.

where $G_\sigma(\cdot)$ represents the Gaussian kernel function, and $\widetilde{\mathbf{M}}$ is the resulting standardized soft mask. In our implementation, the standard deviation σ is dynamically determined as a proportion of the proposal's geometric scale to adaptively balance noise suppression and feature retention. The obtained standardized soft mask $\widetilde{\mathbf{M}} \in [0,1]^{H\times W}$ has values close to 1 at the center of the foreground and gradually decreases toward the boundaries. $\widetilde{\mathbf{M}}$ is used to perform element-level reweighting of the foreground feature $\mathbf{F}_{\text{ins}}^{\text{ob}}$, thereby generating weighted features with spatial awareness. Through soft-weighted global average pooling, the prototype vector $\mathbf{P} \in \mathbb{R}^D$ of the instance is obtained

$$\mathbf{P} = \frac{\sum_{i=1}^{L} \widetilde{m}_i \cdot \mathbf{f}_i}{\sum_{i=1}^{L} \widetilde{m}_i + \epsilon_s} \tag{9}$$

where $\epsilon_s = 10^{-6}$ ensures numerical stability, $\widetilde{m}_i$ is the soft weight scalar from $\widetilde{\mathbf{M}}$ at spatial location i, and $\mathbf{f}_i$ is the feature at the ith spatial location of $\mathbf{F}_{\text{ins}}^{\text{ob}}$.

Repeat this process for all the supporting samples, and a set of initial instance prototypes $\mathcal{P} \in \mathbb{R}^{N\times K\times D}$ will be obtained, which forms the basis for further optimization. To enhance intraclass semantic consistency and suppress potential outliers, we regard $\mathcal{P}$ as the intermediate input of a momentum-based multi-instance learning mechanism. Specifically, we maintain a set of $\mathcal{C}$ updateable prototype centers for each category in the momentum queue to stabilize the training process. Then, we apply online clustering to iteratively update these $\mathcal{C}$ centers using the initial prototypes $\mathcal{P}$. This process generates the final enhanced multi-instance representation: $\widetilde{\mathcal{F}}_{\text{ins}}^{\text{ob}} \in \mathbb{R}^{N\times C\times D}$, where C represents the prototype cardinality assigned to each category, and in our experimental configuration, it aligns with the clustering quantity $C = K$. This design ensures that each supporting sample contributes to a specific, noise-resistant prototype center, thereby significantly improving the robustness of the representation. The SMPA module achieves precise emphasis on the foreground region and smooth attenuation near the boundaries through its continuous and differentiable soft mask. Importantly, this module does not contain learnable parameters, has high computational efficiency, and integrates naturally into the end-to-end training pipeline.

E. PRKA Module

To address domain shift and category confusion in CD-FSOD caused by limited annotations, we propose a knowledge bank-based PRKA module built upon the CD-ViTO baseline. Unlike conventional affinity-matrix-based similarity, PRKA employs an attention mechanism for fine-grained, differentiable knowledge transfer. While sharing the goal of mitigating information scarcity with existing feature recycling methods like the Feature Reuse Strategy in HSIC-FM [46], PRKA fundamentally differs in scope and mechanism. Specifically, HSIC-FM relies on intradomain, pixel-level feature concatenation across network layers to prevent information decay. In contrast, PRKA focuses on interdomain, instance-level knowledge transfer, utilizing an external dynamic key–value cache bank and cross-attention to semantically align cross-domain samples and explicitly address domain shift. As illustrated in Fig. 4, PRKA interacts with an external cache bank $\mathcal{M}$ (acting as the feature repository) to retrieve and index high-fidelity instance representations from the target domain. This mechanism is instrumental in fortifying the model against precision erosion induced by domain-specific distribution shifts. Specifically, the PRKA module extracts ROI-aligned query features $\mathbf{F}_{q_{\text{roi}}}$, which are processed through a patch-based cross-attention mechanism using matrix multiplication with the cached keys and values. Furthermore, this module implicitly achieves semantic-level foreground alignment. By retrieving and adaptively fusing robust source-domain prototypes that are semantically similar to the target queries, PRKA compensates for the lack of target data and aligns the semantic distribution of target objects with the source domain.

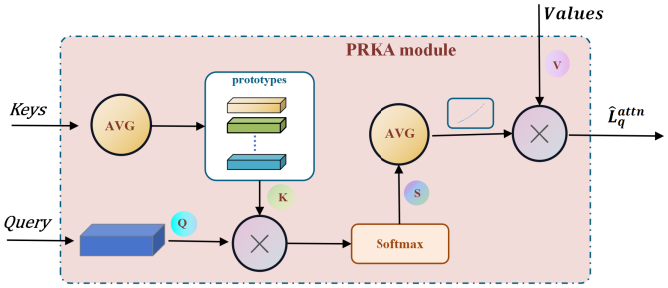


Fig. 4. Detailed framework of PRKA, illustrating cross-domain knowledge retrieval via query–key–value (QKV) attention with the cache bank.

The original CD-ViTO classification head projects ROI-aligned query features $\mathbf{F}_{q_{\text{roi}}} \in \mathbb{R}^{B\times D\times H\times W}$ (with B proposals, feature dimension D, and spatial size $H \times W = 7 \times 7$) into class logits via global average pooling followed by a linear

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classifier $W_T^{\text{cls}} \in \mathbb{R}^{D\times N}$, yielding $\mathbf{F}_{q_{\text{roi}}} W_T^{\text{cls}} \in \mathbb{R}^{B\times N}$. However, it lacks access to historical cross-domain knowledge.

PRKA overcomes this by fusing prototype-derived knowledge through a residual connection. First, a key–value cache bank $\mathcal{M}$ is constructed using enhanced instance prototypes from the SMPA module (Section III-D)

$$\mathbf{E}_F = \text{OneHot}(\mathcal{C}_N) \in \mathbb{R}^{(N\times K)\times N} \tag{10}$$

$$\mathcal{M} = \left\{\text{Key} = F_{\text{ins}}^{\tilde{\text{ob}}},\ \text{Value} = \mathbf{E}_F\right\} \tag{11}$$

where $\mathcal{C}_N = \{1,\ldots,N\}$ represents the set of novel category indices, N is the number of novel classes, and K is the number of support examples for each class. The keys $F_{\text{ins}}^{\tilde{\text{ob}}} \in \mathbb{R}^{N\times K\times D}$ are foreground instance features, and the values $\mathbf{E}_F$ are their one-hot semantic embeddings.

During inference, for each query image, we first extract the region features $F_{q_{\text{roi}}}$. As shown in Fig. 4, $F_{q_{\text{roi}}}$ serves as the query (Q), while the features from the cache bank provide the keys (K) and values (V). We introduce a patch-based cross-attention mechanism to aggregate external knowledge from the cache bank. $F_{q_{\text{roi}}}$ is flattened into a set of query vectors $Q \in \mathbb{R}^{B\times N_{\text{patch}}\times D}$ ($N_{\text{patch}} = H \times W$), which interacts with the cached instance keys $M_{\text{key}} \in \mathbb{R}^{N_p\times D}$, where $N_p = N \times K$ denotes the total number of prototypes, formed by N novel categories with K shots each. The similarity scores are computed via matrix multiplication between Q and M_{key} (indicated by $\otimes$ in the figure), and the patch-level attention weights $\mathbf{A} \in \mathbb{R}^{B\times N_{\text{patch}}\times N_p}$ are obtained after Softmax normalization

$$\mathbf{A} = \text{Softmax}\left(\frac{QM_{\text{key}}^{\top}}{\sqrt{D}}\right) \tag{12}$$

where $\sqrt{D}$ is the scaling factor to prevent gradient vanishing during dot-product attention. The ROI-level instance response $\mathbf{r} \in \mathbb{R}^{B\times N_p}$ is obtained by averaging the attention weights across the spatial dimension

$$\mathbf{r} = \frac{1}{N_{\text{patch}}}\sum_{i=1}^{N_{\text{patch}}} \mathbf{A}_i. \tag{13}$$

The enhanced logits $\hat{L}_q^{\text{attn}} \in \mathbb{R}^{B\times N}$ are derived using the response attenuation mechanism, which corresponds to the matrix multiplication between the response vector and the cache bank values (M_{value}) in Fig. 4. By aggregating the cache bank values $M_{\text{value}} \in \mathbb{R}^{N_p\times N}$, we have

$$\hat{L}_q^{\text{attn}} = \text{Softmax}\left(\exp(\mathbf{r}-1)\times M_{\text{value}}\right) \tag{14}$$

where $\times$ denotes matrix multiplication, and $\exp(\mathbf{r}-1)$ serves as a nonlinear attenuation factor. Since the attention response $\mathbf{r}$ lies in $[0,1]$, the term $(\mathbf{r}-1)$ is negative, and the exponential function maps it to $[1/e, 1]$. This design imposes a "soft gate" that heavily penalizes low-confidence retrievals (where $\mathbf{r} \approx 0$) while preserving high-confidence matches, effectively suppressing irrelevant noise from the cache bank. Conversely, the term $F_{q_{\text{roi}}} W_T^{\text{cls}}$ represents the prediction based on the current linear classifier, which relies solely on the ROI features of the current query. Final prediction $\hat{f}_q$ is obtained by linearly combining the enhanced logits with the original classifier output $F_{q_{\text{roi}}} W_T^{\text{cls}}$ (residual fusion, $\gamma = 0.3$, see Section IV-H)

$$\hat{f}_q = \gamma \hat{L}_q^{\text{attn}} + (1-\gamma) F_{q_{\text{roi}}} W_T^{\text{cls}} \tag{15}$$

where γ is the fusion weight balancing the contributions of the cache bank and the base classifier.

The PRKA module addresses domain shift and category confusion in CD-FSOD by introducing a patch-based cross-attention mechanism. It dynamically aggregates fine-grained prototype knowledge from a cache bank, calculating enhanced logits $\hat{L}_q^{\text{attn}}$ via a response attenuation function. The final prediction is achieved by residual fusion ($\hat{f}_q$) of the enhanced logits with the original classifier output. This design enables efficient and plug-and-play integration and significantly improves accuracy even in extremely small sample sizes.

IV. Experiments

A. Datasets

To evaluate the proposed framework in various remote sensing scenarios, we conducted experiments on six representative datasets, systematically assessing the domain adaptability and generalization ability of BAPA-Net under different cross-domain settings. Following the experimental paradigm established by DE-ViT [38] and CD-ViTO [10], the model was initialized through MS-COCO pretraining and performed specific adaptations and performance benchmark tests for the tasks in a series of heterogeneous target scenarios (including high-altitude perspectives, uniformity of planetary surfaces, underwater environments with signal attenuation, and cross-modal changes).

MS-COCO dataset is a comprehensive benchmark for general object localization. It contains a large training sample library (118 000) and a validation sample library (5000), covering 80 common semantic categories. It provides over 860 000 fine-grained instance-level annotations, with dense supervision signals, which are helpful for achieving robust feature initialization. Each instance includes an axis-aligned bounding box and pixel-level segmentation masks. Its complex background, multiscale objects, and severe occlusions make it the default source domain for domain generalization research. We use it to pretrain the basic detector so that the knowledge can be transferred to the remote sensing domain subsequently.

DIOR [47] contains 23 463 satellite images from Google Earth and high-resolution satellites (each image size is 800 × 800), covering 20 geographical spatial categories (such as aircraft, bridges, ports, and storage tanks) with 192 442 labeled instances. Due to significant differences in scale, density, and background clutter, DIOR has become the main target domain. We perform K rounds of fine-tuning according to the method of CD-ViTO [10] and report the results on its official validation set (2346 images).

NWPU VHR-10 [48] contains 800 target object samples and 100 high spatial resolution samples only containing backgrounds, with ground sampling distances ranging from 0.5 to 2 m. This benchmark data comes from Google Earth and Waihenggen and covers ten different geographical spatial

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categories, including sea vessels, transportation infrastructure, and sports facilities. Its size is moderate, and the annotations are clear, making it widely used for small samples and cross-domain validation. We use it as an auxiliary target domain to further test the generalization ability.

UAV [49] contains 1359 drone images from urban environments captured from ground or aerial perspectives, including bounding boxes of small multiposes drones under different lighting conditions. This dataset poses substantial difficulty, largely because targets are small, backgrounds vary over time, and viewpoints are highly diverse. We use it to evaluate the robustness of BAPA-Net in dealing with airborne moving targets.

MLCD [50] (Mars and Moon Crater Detection Dataset) contains 2000 high-resolution planetary images from NASA's orbiters, with approximately 5000 craters labeled. Due to a uniform background, sparse targets, and the lack of obvious semantic cues, MLCD exhibits extreme domain differences. It is used to evaluate the generalization ability in non-Earth and nonatmospheric conditions.

Underwater Object Detection Dataset (UODD) [10] originates from URPC and contains 2942 real underwater farm images. This dataset poses substantial difficulty, largely because targets are small, backgrounds vary over time, and viewpoints are highly diverse. These images have common underwater visual problems such as color bias (predominantly blue-green), low contrast, motion blur, and suspended particles.

SSDD [51] aims to achieve cross-modal ship detection, including registration data of over 2000 optical images and SAR images under various sea conditions. We adopt SSDD to assess cross-modality transferability under strongly heterogeneous imaging conditions.

To ensure the rigor and objectivity of the benchmark test, our experimental framework adheres to the fine-tuning principles stipulated in CD-ViTO [10]. We evaluate it through the average value of the average precision (mAP@0.5:0.95), which is a quantitative metric. This metric has been systematically evaluated across the entire low-data-volume scenarios (1/5/10 exposures) covering six datasets.

B. Evaluation Metrics

We report mAP_{50}, defined as the mean AP at an IoU threshold of 0.5. Our primary metric is COCO-style mAP, which averages AP over ten IoU thresholds uniformly sampled from $[0.5, 0.95]$ with a stride of 0.05. For each class, AP is computed as the area under the precision–recall curve $P(R)$

$$AP = \int_0^1 P(R)\, dR \tag{16}$$

where [P(R)] denotes the precision–recall (PR) curve, which characterizes how precision (P) varies as a function of recall (R) over the full operating range. Mean average precision (AP) summarizes detector performance by averaging the per-class AP scores across all object categories

$$mAP = \frac{1}{C} \sum_{c=1}^{C} AP_c \tag{17}$$

where C represents the cardinality of the semantic category set, while AP_c represents the precision indicator specific to the cth indexed category.

C. Implementation Details

We implement BAPA-Net on top of Detectron2 and train it on a single NVIDIA RTX A5000 GPU equipped with 24 GB of VRAM. We base the architecture on the design of DE-ViT [38] and adopt a dual backbone network strategy. By initializing ResNet-50 with ImageNet and combining the visual transformer (ViT) based on DINOv2 for prealignment in the MS-COCO domain, we evaluate the framework in different scenarios with $K \in \{1, 5, 10\}$ following the standard N-class K-times paradigm [10]. The internal dynamics of the model are controlled by a set of globally consistent hyperparameters: specifically, the CBFA module uses $\beta = 0.4$ and $\varepsilon = 0.2$ to regulate entropy and alignment, while the PRKA module combines residuals with $\gamma = 0.3$, as detailed in Section IV-H. We optimize the model using SGD with a batch size of 16 and an initial learning rate of 2×10^{-3}. For a single task, the training runs for 80 epochs, whereas the 5- and 10-adaptation settings each converge with 40 epochs.

D. CD-FSOD Results in Remote Sensing Scenarios

Table I presents the performance evaluation results of BAPA-Net compared with existing SOTA methods. This evaluation covers six different remote sensing benchmark tests, ranging from high-altitude multiscale geographical spatial scenes (DIOR and NWPU VHR-10), low-altitude scenes (UAV), deep-space scenes (MLCD), to specialized underwater and planetary domains (UODD, SSDD). The performance of our framework is quantified through the strict mAP@0.5:0.95 evaluation metric under low data conditions with 1, 5, and 10 data collection times. It should be noted that we did not directly compare with recent open-source methods such as Domain-RAG [41], as their evaluation protocols are not as strict. Specifically, their source-domain pretraining usually includes heterogeneous external datasets, which may unintentionally contain statistical information of the target domain, thereby introducing potential data leakage and undermining the fairness of cross-domain generalization evaluation.

From the results, BAPA-Net significantly outperforms all competing methods under most few-shot settings and mainstream remote sensing scenarios, demonstrating excellent generalization ability and adaptability, especially in scenarios with extremely limited data. In the 1-shot setting, the average mAP of BAPA-Net reaches 15.4%, significantly surpassing the previous best performer CD-ViTO [10] (13.9%), and achieving new SOTA results on DIOR, NWPU VHR-10, UAV, and SSDD. This result suggests that the model can recover discriminative semantic and spatial representations even with minimal supervision, indicating strong robustness to domain shifts.

As the number of support sets gradually increases, BAPA-Net can still effectively scale up, with its average mAP scores reaching 22.8 (for 5 samples) and 27.1 (for 10 samples), respectively, outperforming CD-ViTO (21.3 and 25.8) and

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TABLE I

PERFORMANCE COMPARISON OF MAP@0.5:0.95 (%) ACROSS SIX REMOTE SENSING BENCHMARKS. RESULTS ARE AVERAGED OVER FIVE STOCHASTIC TRIALS (MEAN ± STD). OPTIMAL AND SUBOPTIMAL SCORES ARE HIGHLIGHTED WITH RED AND BLUE TRANSLUCENT OVERLAYS. THE "AVERAGE" COLUMN REPORTS THE MEAN MAP ACROSS ALL TESTED DOMAINS TO REFLECT THE OVERALL CROSS-DOMAIN GENERALIZATION CAPABILITY, AND "†" INDICATES OUR CUSTOMIZED IMPLEMENTATIONS OR LOCALLY REPRODUCED RESULTS

Shot	Method	Backbone	mAP (%)						
			DIOR	NWPU VHR-10	UAV	MLCD	UODD	SSDD	Average
1-shot	Meta-RCNN [12]	ResNet50	7.80±0.25	—	—	—	3.60±0.18	—	5.70±0.21
	TFA w/cos [52]	ResNet50	8.00±0.27	—	—	—	4.40±0.20	—	6.20±0.23
	FSCE [53]	ResNet50	8.60±0.29	—	—	—	3.90±0.19	—	6.30±0.24
	DeFRCN [54]	ResNet50	9.30±0.31	—	—	—	4.50±0.22	—	6.90±0.26
	Distill-cdfsod [55]	ResNet50	10.50±0.35	9.60±0.33	11.00±0.30	2.80±0.15	5.90±0.28	15.30±0.40	9.20±0.32
	ViTDet-FT† [56]	ViT-B/14	12.90±0.38	11.30±0.36	12.30±0.32	3.00±0.16	4.00±0.21	8.60±0.30	8.70±0.31
	Detic [57]	ViT-L/14	0.10±0.02	0.10±0.02	2.30±0.12	—	—	—	0.40±0.05
	Detic-FT † [57]	ViT-L/14	4.10±0.20	3.80±0.19	5.30±0.14	3.20±0.17	4.20±0.21	3.60±0.18	4.00±0.19
	DE-ViT [38]	ViT-L/14	2.70±0.15	2.30±0.13	4.60±0.11	2.00±0.12	1.50±0.09	2.90±0.16	2.70±0.14
	DE-ViT-FT† [38]	ViT-L/14	14.70±0.42	13.60±0.40	16.60±0.28	3.20±0.17	2.40±0.14	11.10±0.35	10.30±0.33
	CD-ViTO [10]	ViT-L/14	17.80±0.32	19.90±0.41	20.80±0.28	6.10±0.19	3.10±0.15	15.70±0.35	13.90±0.24
	BAPA-Net (Ours)	ViT-L/14	18.20±0.29	21.40±0.37	24.20±0.43	7.20±0.22	4.70±0.18	16.60±0.31	15.40±0.26
5-shot	Meta-RCNN [12]	ResNet50	17.70±0.45	—	—	—	8.80±0.32	—	13.30±0.40
	TFA w/cos [52]	ResNet50	18.10±0.46	—	—	—	8.70±0.31	—	13.40±0.41
	FSCE [53]	ResNet50	18.70±0.48	—	—	—	9.60±0.34	—	14.20±0.44
	DeFRCN [54]	ResNet50	18.90±0.49	—	—	—	9.90±0.35	—	14.40±0.45
	Distill-cdfsod [55]	ResNet50	19.10±0.50	17.90±0.47	25.30±0.58	6.80±0.28	12.20±0.42	16.50±0.46	16.30±0.48
	ViTDet-FT† [56]	ViT-B/14	23.30±0.55	22.10±0.52	27.90±0.60	7.30±0.30	11.10±0.38	12.20±0.42	17.30±0.50
	Detic [57]	ViT-L/14	0.10±0.02	0.10±0.02	10.30±0.36	—	—	—	1.80±0.10
	Detic-FT† [57]	ViT-L/14	12.10±0.40	11.60±0.39	26.40±0.59	4.10±0.22	10.40±0.37	6.50±0.28	11.90±0.42
	DE-ViT [38]	ViT-L/14	7.80±0.30	6.50±0.27	12.30±0.38	2.50±0.15	3.10±0.18	4.10±0.22	6.10±0.26
	DE-ViT-FT† [38]	ViT-L/14	21.20±0.52	20.30±0.50	36.80±0.65	8.20±0.33	5.00±0.25	12.30±0.42	17.30±0.51
	CD-ViTO [10]	ViT-L/14	26.90±0.45	26.30±0.52	41.60±0.61	9.60±0.27	6.80±0.21	16.70±0.38	21.30±0.33
	BAPA-Net (Ours)	ViT-L/14	27.80±0.41	29.20±0.48	43.00±0.57	12.00±0.30	7.30±0.23	17.50±0.34	22.80±0.29
10-shot	Meta-RCNN [12]	ResNet50	20.60±0.52	—	—	—	11.20±0.38	—	15.90±0.46
	TFA w/cos [52]	ResNet50	20.50±0.51	—	—	—	11.80±0.40	—	16.20±0.47
	FSCE [53]	ResNet50	21.90±0.54	—	—	—	12.00±0.41	—	17.00±0.49
	DeFRCN [54]	ResNet50	22.90±0.56	—	—	—	12.10±0.42	—	17.50±0.51
	Distill-cdfsod [55]	ResNet50	26.50±0.60	27.30±0.62	38.70±0.68	9.80±0.35	14.50±0.48	16.30±0.46	22.20±0.55
	ViTDet-FT† [56]	ViT-B/14	29.40±0.65	30.50±0.67	42.60±0.72	10.80±0.38	15.60±0.50	12.20±0.42	23.50±0.58
	Detic [57]	ViT-L/14	0.10±0.02	0.10±0.02	13.60±0.42	—	—	—	2.30±0.12
	Detic-FT† [57]	ViT-L/14	15.40±0.48	16.30±0.50	20.80±0.52	6.40±0.28	14.40±0.47	6.90±0.30	13.40±0.45
	DE-ViT [38]	ViT-L/14	8.40±0.32	9.20±0.34	16.50±0.45	3.50±0.20	3.10±0.18	4.70±0.24	7.60±0.30
	DE-ViT-FT† [38]	ViT-L/14	25.60±0.59	28.30±0.63	42.60±0.72	9.50±0.34	5.40±0.26	12.50±0.43	20.70±0.54
	CD-ViTO [10]	ViT-L/14	30.80±0.51	34.10±0.59	53.40±0.72	12.50±0.33	7.00±0.25	17.00±0.40	25.80±0.37
	BAPA-Net (Ours)	ViT-L/14	31.40±0.47	35.20±0.54	54.00±0.68	14.60±0.36	7.90±0.27	19.70±0.38	27.10±0.32

ViTDet-FT [56] (17.3 and 20.7). Notably, on the challenging UAV and MLCD datasets, BAPA-Net achieved excellent mAP scores of 43.0 and 12.0 in the 5-sample setting, and further improved to 54.0 and 14.6 in the 10-sample setting, significantly surpassing all benchmark models. The observed gains stem from the synergistic interaction of three core modules, which together address two primary bottlenecks encountered in low-shot settings: domain differences and prototype ambiguity.

It is worth noting that while BAPA-Net achieves robust baseline performance in typical aerial and satellite remote sensing scenarios, it faces greater challenges in highly difficult underwater scenes with extreme optical distortions, leading to a less pronounced improvement on the UODD dataset that is notably lower than the methods proposed by Distill-cdfsod [55] and ViTDet-FT† [56]. Severe underwater optical degradation considerably disrupts the shared visual structure between the source domain (MS COCO) and the underwater target domain. Meanwhile, our CBFA module, though effective in alleviating common geospatial and structural domain shifts, cannot be fully leveraged under such extreme nonlinear optical degradation caused by water turbidity and light propagation. Nevertheless, the proposed framework maintains favorable stability across all scenarios, and these results highlight that extreme cross-domain transfer in underwater environments remains a challenging frontier topic that warrants further exploration.

E. Qualitative Visualization of CD-FSOD in Remote Sensing Scenarios

To further assess the practical efficacy of our framework, Fig. 5 conducts a qualitative comparison of the performance of DE-ViT-FT [38] (the fine-tuned version of DE-ViT), CD-ViTO [10], and the proposed BAPA-Net on representative cross-domain remote sensing scenarios

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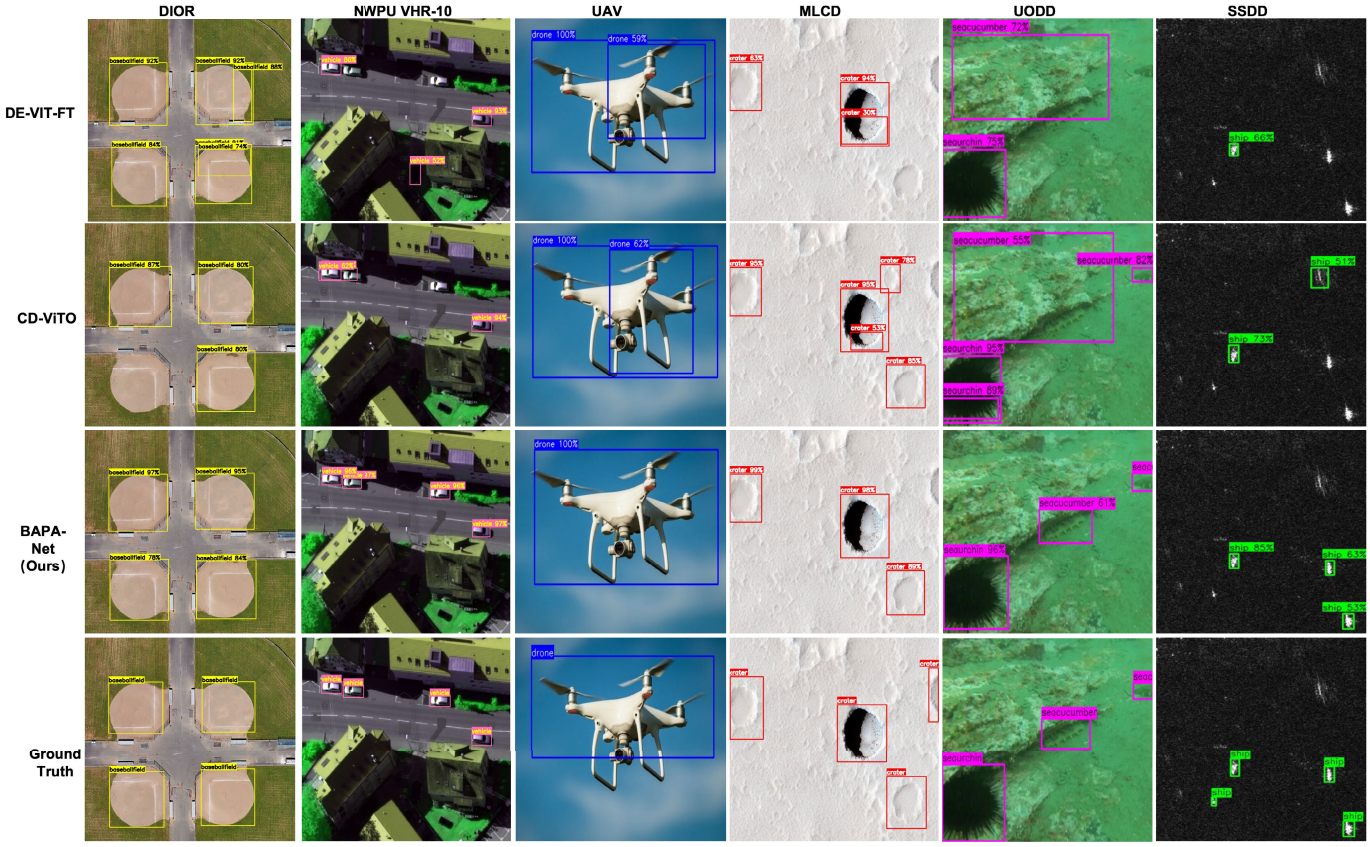


Fig. 5. Visualization of 5-shot detection results by DE-ViT-FT, CD-ViTO, and BAPA-Net on diverse remote sensing datasets.

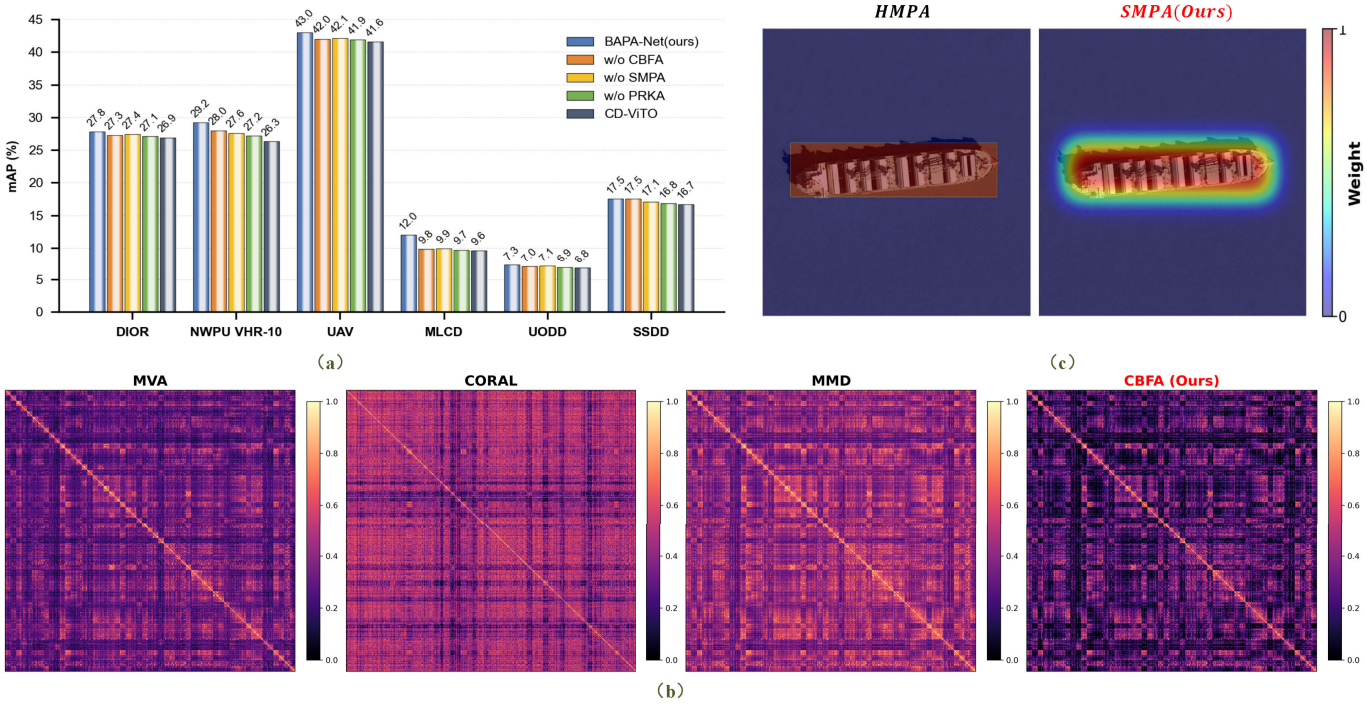


Fig. 6. Quantitative and visual ablation of BAPA-Net components (5-shot). (a) Componentwise ablation study on six remote sensing datasets (quantitative). (b) Visualization of cosine similarity matrices of fused background prototypes under different fusion strategies, demonstrating the intraclass compactness and interclass discriminability of CBFA (Ours). (c) Visual comparison between HMPA and SMPA (Ours, showing the more precise and adaptive weight distribution of SMPA.

(including DIOR, NWPU VHR-10, UAV, MLCD, UODD, and SSDD). All results are compared with the manually annotated benchmark data (GT) to visually highlight the performance differences.

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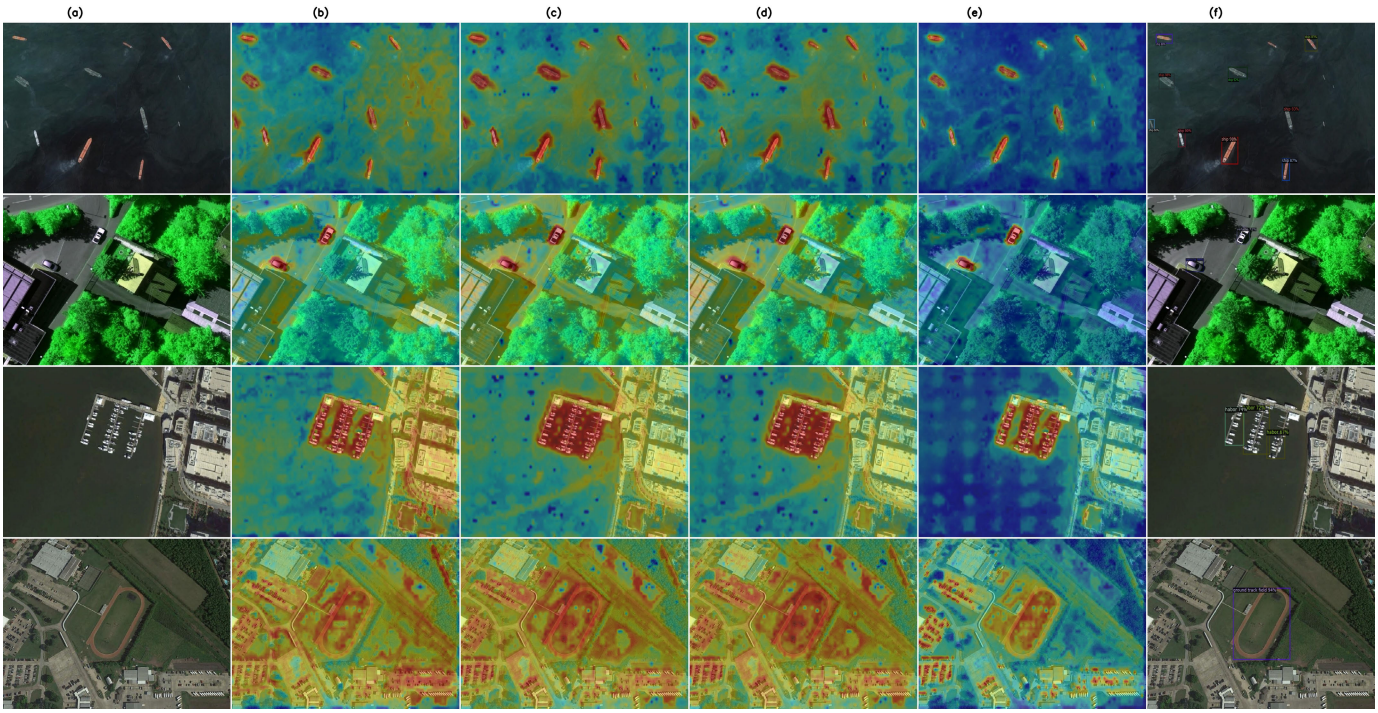


Fig. 7. Visual ablation of feature maps and CAMs on NWPU VHR-10 (5-shot): (a) Original input image. (b) Baseline (without proposed modules), showing scattered activations due to background interference. (c) +CBFA module, demonstrating effective background noise suppression. (d) +SMPA module, exhibiting sharper focus on target objects by attenuating peripheral clutter. (e) BAPA-Net (full model with PRKA), achieving the strongest and most complete object activation. (f) Final detection results predicted by BAPA-Net.

TABLE II

PERFORMANCE COMPARISON BETWEEN SINGLE-DOMAIN FSOD AND OUR CROSS-DOMAIN FSOD FRAMEWORK (BAPA-NET) ON THE DIOR DATASET (MAP%). RED/BLUE SHADING DENOTES THE BEST AND RUNNER-UP RESULTS PER COLUMN

Method	Setting	3-shot	5-shot	10-shot	20-shot
Meta-RCNN [12]	RS-FSOD	8.8	10.9	14.9	16.7
TFA w/cos [52]	RS-FSOD	8.1	8.2	8.7	12.1
PCNN [58]	RS-FSOD	14.5	15.0	18.9	22.8
G-FSOD [7]	RS-FSOD	14.1	15.8	20.7	22.7
SAE-FSDet [59]	RS-FSOD	14.0	15.7	17.4	21.3
GE-FSOD [60]	RS-FSOD	14.5	16.7	19.0	26.1
ST-FSOD [6]	RS-FSOD	17.7	20.7	27.3	33.4
PL-FSOD [5]	RS-FSOD	18.1	21.1	28.0	34.1
SMDC [61]	RS-FSOD	18.4	21.8	25.4	30.0
BAPA-Net (Ours)	CD-FSOD	20.8	27.8	31.4	35.6

TABLE III

COMPARISON OF CBFA WITH CLASSICAL BACKGROUND AND STATISTICAL ALIGNMENT METHODS ON DIOR AND NWPU VHR-10 (MAP%). ARROWS INDICATE PERFORMANCE CHANGES RELATIVE TO CBFA, AND RED SHADED CELLS INDICATE THE BEST PERFORMANCE

Fusion Method				DIOR		NWPU10	
MVA	CORAL	MMD	CBFA	1-shot	5-shot	1-shot	5-shot
✓	×	×	×	$18.0^{\downarrow 0.2}$	$26.7^{\downarrow 1.1}$	$21.2^{\downarrow 0.2}$	$29.2^{\downarrow 0.0}$
×	✓	×	×	$14.4^{\downarrow 3.8}$	$23.8^{\downarrow 4.0}$	$14.9^{\downarrow 6.5}$	$25.9^{\downarrow 3.3}$
×	×	✓	×	$18.1^{\downarrow 0.1}$	$27.6^{\downarrow 0.2}$	$21.1^{\downarrow 0.3}$	$28.2^{\downarrow 1.0}$
×	×	×	✓	**18.2**	**27.8**	**21.4**	**29.2**

As shown in Fig. 5, BAPA-Net produces cleaner detections in cluttered remote sensing scenes. DE-ViT-FT and CD-ViTO more often confuse targets with background textures, which leads to missed detections, false alarms, and imprecise boxes, while BAPA-Net reduces these errors. In the DIOR baseball field example, BAPA-Net localizes all four fields and preserves thin boundary structures; CD-ViTO misses one field, and both baselines blur the boundary lines. In the dense NWPU VHR-10 building scene, the baselines generate more erroneous predictions under cluttered backgrounds, whereas CBFA helps BAPA-Net separate vehicles from shadow regions. Across other general remote sensing scenarios (UAV and MLCD), the method remains stable under large-scale variations and subpixel targets through prototype retrieval and feature purification. In more challenging settings, the improvements are smaller but still consistent. In the underwater UODD domain, multisource aggregation reduces confusion between similar objects and complex backgrounds. In the cross-modal SSDD scenario, BAPA-Net recovers more small targets despite SAR speckle noise. Some extremely tiny objects are still missed, but these results support the combined effect of the CBFA, SMPA, and PRKA modules. To examine what drives these improvements, Section IV-G (Figs. 6 and 7) provides feature-level visualizations that illustrate how our modules align backgrounds and suppress clutter.

F. FSOD on Remote Sensing Benchmarks

To clarify the performance gap and practical value between cross-domain and traditional single-domain object detection methods, Table II presents a comparative study of the SOTA single-domain object detection methods and our cross-domain method (BAPA-Net) under sample settings of 3/5/10/20. Conventional single-domain detectors typically depend on extensive base-class annotations collected in the target domain to establish domain-specific prior knowledge before fine-tuning, and these basic class annotations are completed during

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the secondary pretraining stage in the target domain. In contrast, BAPA-Net adopts a closed-world evaluation protocol that completely eliminates target-domain base-class pretraining. This framework operates by leveraging an inheritance from a source-pretrained backbone, executing knowledge transfer exclusively through cross-domain adaptation using a nominal set of labeled exemplars from the target-specific novel categories.

Experimental results show that, under identical shot constraints, BAPA-Net achieves an average mAP of 28.9—despite having no access to target-domain base-class data—surpassing the competitive single-domain method (PL-FSOD, 25.3) by 3.6% points. The advantage is particularly pronounced in moderate-to-high shot regimes (5-shot and above). Even in the extreme 3-shot setting, BAPA-Net improves mAP by 2.4%points (20.8 versus 18.4), demonstrating the efficacy of our cross-domain knowledge transfer mechanism.

From a labeling-cost perspective, single-domain FSOD demands hundreds to thousands of annotated base-class images in the target domain, whereas BAPA-Net requires only a handful of novel-class annotations to achieve high performance. This drastically reduces the human and temporal costs of expert annotation in remote sensing applications, better addressing the real-world challenge of scarce and cross-domain distributed labels.

G. Ablation Study

To quantify the individual impact of our integrated components, we conduct a systematic ablation study by isolating the effects of CBFA, SMPA, and PRKA. Quantitative results across benchmark tests are summarized in Fig. 6(a) and Table III, while Figs. 6(b) and (c) and 7 provide visual evidence of background suppression, prototype purification, and feature activation refinement, respectively.

1) Componentwise Ablation Study: Fig. 6(a) summarizes the ablation study of BAPA-Net under the 5-shot protocol on six remote sensing benchmarks, where we adopt a Leave-One-Out strategy—systematically removing one component at a time from the full framework—to quantify its individual contribution and indispensability. Using the *NWPU VHR-10* dataset as a representative case, the complete BAPA-Net achieves 29.2% mAP. The results reveal a nonlinear synergistic effect between our modules: removing CBFA, SMPA, or PRKA individually leads to performance drops of 1.2%, 1.6%, and 2.0%, respectively. Notably, the sum of these individual drops (4.8%) exceeds the net gain over the baseline (2.9%), confirming that our components are mutually reinforcing; for instance, the SMPA module's foreground purification is significantly more effective when the CBFA module has already stabilized the background distribution. Finally, removing all components results in a baseline performance of 26.3%, essentially matching the CD-ViTO baseline. Beyond this example, the same qualitative pattern is consistently observed across the remaining benchmarks in Fig. 6(a): each module yields a non-negligible improvement, and their integration produces the most favorable accuracy, supporting the effectiveness and general applicability of the proposed design in diverse remote sensing scenarios.

2) Comparison of Background Fusion Methods: To rigorously validate the design rationale of the CBFA module, we benchmark it against classical domain adaptation baselines, including mean-variance alignment (MVA), CORAL (correlation alignment), and maximum mean discrepancy (MMD). As detailed in Table III, although statistical moment-matching methods (MVA and CORAL) offer marginal improvements by aligning global feature distributions, they struggle to model the complex, nonstationary background variations typical in remote sensing imagery. In contrast, CBFA leverages optimal transport to construct a geometrically meaningful correspondence between source and target background prototypes, yielding a superior mAP of 27.80% on DIOR (5-shot). This represents a significant margin over MVA (+ 1.10%) and MMD (+ 0.22%), confirming that the performance gain stems from the precise structural alignment of background semantics rather than mere statistical regularization. Fig. 6(b) presents the cosine similarity matrices of fused background prototypes under different fusion strategies. CBFA (Ours) exhibits the clearest block structure with high intraclass similarity (bright red) and low interclass similarity (dark blue), demonstrating strong pattern discriminability and avoiding the oversmoothing issue of CORAL and MMD. The sparse low-similarity distribution of CBFA (Ours) further verifies the high purity of the fused prototypes, which directly contributes to the improved detection accuracy in remote sensing scenes.

3) Effectiveness of SMPA: To elucidate the mechanism of prototype purification, Fig. 6(c) provides a visual contrast between hard mask prototype augmentation (HMPA) and our proposed SMPA. Specifically, the left side of Fig. 6(c) visualizes HMPA, which imposes a rigid, binary rectangular window assigning uniform weights (indicated by solid colors) to all pixels within the bounding box. In the context of remote sensing, where objects often exhibit arbitrary orientations and high aspect ratios (e.g., bridges and ships), this rigid masking inevitably captures substantial background clutter, introducing domain-specific noise into the prototype. Conversely, the right side illustrates SMPA, which employs a dynamic Gaussian soft mask. As visually depicted by the fading color gradient, it assigns higher importance (warmer colors) to the semantic core of the object while gradually attenuating the weights (cooler colors) of the peripheral context. This "soft-focus" mechanism effectively decouples the object from its surrounding environment, ensuring that the aggregated prototypes remain robust to background shifts and focus on domain-invariant structural features.

4) Componentwise Feature Visualization: To intuitively demonstrate the impact of each module on feature representation, we visualize the class activation maps (CAMs) of the target object features in Fig. 7.

1) The Baseline model suffers from significant background interference, with activations scattered across irrelevant regions.
2) Upon introducing the CBFA module, the background noise is noticeably suppressed, indicating that

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cross-domain background alignment effectively filters out domain-specific distractors.

3) With the addition of the SMPA module, the focus on the target object becomes much sharper, as the soft-mask mechanism refines the prototype by attenuating peripheral clutter.
4) Finally, the complete BAPA-Net (incorporating PRKA) exhibits the strongest and most complete activation on the object of interest, demonstrating that retrieved prototypes provide critical guidance for precise localization. These visualizations corroborate the quantitative improvements observed in Fig. 6(a) and confirm that each component contributes distinctively to enhancing feature discriminability.

The experimental results in Table III, Fig. 6, and Fig. 7 collectively demonstrate that all components of BAPA-Net can work together effectively. They can enhance the feature alignment effect, improve the discrimination ability of the prototypes, and enhance the final detection performance, while maintaining extremely high computational efficiency.

H. Hyperparameter Sensitivity Analysis

To quantitatively evaluate the influence and stability of the key hyperparameters in BAPA-Net, we employ a univariate sensitivity analysis (One-at-a-Time) strategy. Instead of an exhaustive grid search, which is computationally prohibitive, we fix the best performing configuration ($\beta = 0.4, \varepsilon = 0.2, \gamma = 0.3$) as the baseline and vary one hyperparameter at a time. Table IV reports the performance impact of perturbing each parameter around this optimum across six remote sensing datasets. Specifically, under the 5-shot setting, we investigate three key hyperparameters: 1) the target-domain background prototype fusion ratio β in the CBFA module; 2) the entropy regularization coefficient ε; and 3) the residual weighting factor γ in the PRKA module. The changes in mAP for each parameter variation on six remote sensing datasets were reported.

Table IV presents the results of the fusion ratio β. This ratio balances source-domain background priors against target-domain background cues. Empirically, ($\beta = 0.4$) yields the best performance on nearly all datasets (except SSDD). When β is set too small (e.g., $\beta = 0.3$), the model cannot incorporate sufficient target-domain background information, which weakens its ability to mitigate domain shift. However, if it is set too high ($\beta = 0.5$), the noise in the target domain will become too intense, thereby disrupting the prototypes of the source domain. Therefore, an intermediate fusion method is needed to coordinate the background. For the entropy coefficient ε, the optimal value is 0.2. At 0.1, the transfer plan is difficult to converge, and at 0.3, excessive smoothing will cause the background features to lose their unique details. Finally, for the PRKA weight factor γ (used to balance the old prototype knowledge and the new ROI features), a value of 0.3 yields the best performance. This is in line with the usual practice in knowledge distillation and memory learning, where a weight of approximately 0.3 is typically used to maintain stability and adapt to new data [62], [63].

TABLE IV

HYPERPARAMETER SENSITIVITY ANALYSIS OF BAPA-NET (5-SHOT, MAP%). ARROWS INDICATE PERFORMANCE CHANGES RELATIVE TO THE REFERENCE CONFIGURATION, AND RED SHADED CELLS HIGHLIGHT THE OPTIMAL PERFORMANCE FOR EACH DATASET

β	ε	γ	DIOR	NWPU10	UAV	MLCD	UODD	SSDD
0.3	0.2	0.3	$27.7^{\downarrow 0.1}$	$28.5^{\downarrow 0.7}$	$42.4^{\downarrow 0.6}$	$11.9^{\downarrow 0.1}$	$7.1^{\downarrow 0.2}$	$17.9^{\uparrow 0.4}$
0.4	0.2	0.3	27.8	29.2	43.0	12.0	7.3	17.5
0.5	0.2	0.3	27.8	$27.9^{\downarrow 1.3}$	$42.7^{\downarrow 0.3}$	$11.7^{\downarrow 0.3}$	$7.1^{\downarrow 0.2}$	$16.9^{\downarrow 0.6}$
0.4	0.3	0.3	$27.7^{\downarrow 0.1}$	$28.2^{\downarrow 1.0}$	$43.4^{\uparrow 0.4}$	$11.0^{\downarrow 1.0}$	$6.7^{\downarrow 0.6}$	17.5
0.4	0.1	0.3	$27.6^{\downarrow 0.2}$	$28.0^{\downarrow 1.2}$	$42.9^{\downarrow 0.1}$	$11.5^{\downarrow 0.5}$	$6.9^{\downarrow 0.4}$	$17.6^{\uparrow 0.1}$
0.4	0.2	0.2	$27.7^{\downarrow 0.1}$	$28.1^{\downarrow 1.1}$	$42.6^{\downarrow 0.4}$	$11.6^{\downarrow 0.4}$	$7.0^{\downarrow 0.3}$	$17.4^{\downarrow 0.1}$
0.4	0.2	0.4	27.8	$28.2^{\downarrow 1.0}$	$42.9^{\downarrow 0.1}$	12.0	$7.0^{\downarrow 0.3}$	$17.7^{\uparrow 0.2}$

TABLE V

ASSESSING THE TRADE-OFF BETWEEN COMPUTATIONAL OVERHEAD AND DETECTION PRECISION FOR BAPA-NET COMPONENTS (10-SHOT, NWPU VHR-10). ALL METRICS ARE MEASURED USING A VIT-L/14 BACKBONE ON A SINGLE NVIDIA RTX A5000 GPU

Model Config.	Params (M)	Training Time (s/img)	Inference Time (s/img)	Accuracy (mAP)
Baseline	331.19	0.209	0.3070	34.10
+ CBFA	331.19	0.209	0.3070	34.85
+ SMPA	331.19	0.209	0.3070	34.42
+ PRKA	331.19	0.209	0.3195	34.95
BAPA-Net	**331.19**	**0.209**	**0.3195**	**35.20**
Net Increase	+0M	+0s	+0.0125s	+1.10

Among all three parameters, even if we adjust the setting values by $\pm 50\%$, the variation in mAP is less than 1.5 points. This indicates that BAPA-Net is not highly sensitive to these choices and has stable and reliable performance. Based on this, we only use $\beta = 0.4, \varepsilon = 0.2, \gamma = 0.3$ as our standard settings, as they can achieve excellent results on almost all datasets.

I. Computational Efficiency Study

Table V comprehensively assesses the computational cost and detection accuracy of BAPA-Net. All BAPA-Net configurations maintain a constant parameter size of 331.19M. This stability stems from the design of the CBFA and SMPA modules. They rely on offline prototype preprocessing and numerical operations rather than introducing additional learnable layers. Similarly, the PRKA module manages static cache tensors for operation without adding additional trainable parameters. In terms of training efficiency, the average processing time for a single image remains at 0.209 s (calculated based on every 3.345 s of iteration and a batch size of 16). This consistency in training time confirms that the components of BAPA-Net are completely decoupled from the gradient backpropagation process, without adding additional burdens to the training stage. In the inference stage, the CBFA and SMPA modules have zero additional time overhead due to direct access to precomputed static tensors and no need for real-time calculations, maintaining a baseline delay of 0.3070 s/image. The PRKA module introduces a minor increment of 12.5 ms due to the involvement of lightweight matrix multiplication in the attention mechanism, raising the total inference time to 0.3159 s. Despite the extremely low overhead, BAPA-Net still achieves a 1.10% improvement in mAP

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compared to the baseline model. The experimental results demonstrate that BAPA-Net achieves an ideal balance between running speed and detection accuracy, effectively improving few-shot object detection performance in remote sensing imagery.

J. Discussion

Regarding the role of background in cross-domain adaptation, a key insight from this study is the re-evaluation of background information in cross-domain adaptation. Traditional approaches often treat background as noise to be filtered. In contrast, our results indicate that the background distribution encapsulates domain-specific characteristics that, when aligned via the CBFA module, provide a common statistical substrate for the source and target domains. This "background harmonization" effectively reduces the domain shift perceived by the feature extractor, allowing the foreground objects to be recognized based on their intrinsic semantics rather than being obscured by environmental discrepancies.

In terms of the synergy between prototype purification and knowledge retrieval, the robustness of BAPA-Net in low-shot scenarios is driven by the complementary functions of the SMPA and PRKA modules. Prototype construction in FSL is often plagued by background contamination and limited sample diversity. SMPA mitigates the former by applying a Gaussian-smoothed mask to isolate the object of interest, while PRKA addresses the latter by retrieving semantically similar prototypes from a source-domain cache. This mechanism effectively constructs a "virtual" support set that is both cleaner and more diverse than the limited target samples alone, significantly enhancing the discriminative power of the final object representations.

From the perspective of algorithmic efficiency and deployment viability, BAPA-Net offers a compelling trade-off between accuracy and computational cost. The proposed modules introduce a negligible latency increase (12.5 ms) and require no additional parameter training during the adaptation phase. This efficiency renders the framework highly suitable for on-board processing in remote sensing platforms, such as UAVs and satellites, where power and compute resources are at a premium. Unlike heavy fine-tuning methods, BAPA-Net enables rapid adaptation to new environments with minimal system overhead.

Finally, regarding the limitations in challenging scenarios, while BAPA-Net shows strong transferability on common optical datasets and our proposed modules effectively handle regular domain shifts, its performance improvements remain limited in cross-modal and extreme environments. Specifically, the model achieves only moderate gains on SAR imagery (SSDD) owing to the inherent physical gap between SAR and optical imaging. Moreover, our method does not reach SOTA results on the underwater UODD dataset, since severe optical degradation severely disrupts domain-shared visual structures and prevents our modules from being fully effective. This further suggests that cross-domain detection under extreme physical environments may require priors from physical imaging models, which will be a key focus of our future work. To address such extreme nonlinear domain gaps, future work will incorporate physical degradation modeling and modality-specific feature learning.

V. Conclusion

This article presents BAPA-Net, a novel framework for CD-FSOD in remote sensing. We propose the CBFA module to rigorously address domain shifts. By employing an optimal transport strategy, CBFA aligns the background distributions of source and target domains, effectively minimizing domain-specific noise with negligible computational overhead. To enhance prototype reliability, we introduce the SMPA module, which utilizes a Gaussian soft-masking mechanism to suppress background clutter while preserving critical fine-grained details. Furthermore, the PRKA module is incorporated to facilitate knowledge transfer across domains via a memory-efficient caching system, introducing only a marginal latency of 12.5 ms. Extensive experiments on six remote sensing datasets demonstrate that BAPA-Net significantly outperforms the SOTA CD-ViTO baseline, achieving mAP improvements ranging from 5.0% to 10.8%. While cross-modal adaptation from optical to SAR imagery remains challenging due to inherent physical disparities, BAPA-Net exhibits robust performance in diverse scenarios, including underwater and deep-space environments. These results validate that explicit background modeling and prototype refinement are pivotal for effective domain adaptation. Future work will focus on exploring advanced strategies such as cross-modal style transfer and modality-specific noise filtering to further unlock the potential of SAR scenarios, alongside investigating diffusion-based generative models to address severe environmental degradation.

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